MATH WITH OUMAIMA

FREE INTEGRATION FORMULA SHEET

A quick revision guide • Essential rules + 3 solved examples

1. Basic Integration Rules

Rule	Integral
Constant	$\int k\,dx = kx + C$
Power rule ($n \neq -1$)	$\int x^n\,dx = x^{n+1}/(n+1) + C$
Reciprocal rule	$\int 1/x\,dx = \ln\lvert x\rvert + C$
Exponential	$\int e^x\,dx = e^x + C$
General exponential ($a > 0, a \neq 1$)	$\int a^x\,dx = a^x/\ln(a) + C$
Sine	$\int \sin(x)\,dx = -\cos(x) + C$
Cosine	$\int \cos(x)\,dx = \sin(x) + C$
Secant squared	$\int \sec^2(x)\,dx = \tan(x) + C$

Remember: *For an indefinite integral, always add the constant of integration* ***+ C****.*

2. Useful Properties

Linearity	$\int [f(x) + g(x)]\,dx = \int f(x)\,dx + \int g(x)\,dx$
Constant multiple	$\int k\cdot f(x)\,dx = k\int f(x)\,dx$
Change of variable	If $u = g(x)$, then $du = g'(x)\,dx$. Rewrite the integral using u.

3. Three Solved Examples

Example 1 — Power Rule

Evaluate: $\int (3x^2 - 4x + 2)\,dx$

Integrate each term separately:
$= 3 \cdot x^3/3 - 4 \cdot x^2/2 + 2x + C$
Answer: $x^3 - 2x^2 + 2x + C$

Example 2 — Exponential Integral

Evaluate: $\int (2e^x + 1/x)\,dx$

Use the exponential and reciprocal rules:
$= 2e^x + \ln|x| + C$
Answer: $2e^x + \ln|x| + C$

Example 3 — Change of Variable

Evaluate: $\int 2x(x^2 + 1)^3\,dx$

Let $u = x^2 + 1$, so $du = 2x\,dx$.
Then $\int 2x(x^2 + 1)^3\,dx = \int u^3\,du = u\blacksquare/4 + C$.
Answer: $(x^2 + 1)\blacksquare/4 + C$

Study tip: Differentiate your final answer to check that you recover the original integrand.